



Face And Object Tracking AI SMART ROBOT

ASHWINI KUMAR SINHA

Precision in human-robot interaction depends on the ability to recognise and track human faces along with detailed facial landmarks. This capability underpins emotional analysis and the development of machine learning models grounded in real human behaviour. By analysing facial features, robots can interpret responses and communicate more naturally.

Face tracking further enables real-time interaction and coordinated movement. However, accurately detecting and tracking multiple facial landmarks for expression and emotion analysis remains a challenge. Advanced robotic systems must monitor over 100 facial points in real time and respond accordingly—for example, by synchronising head movements with the user and maintaining natural eye contact.

This system addresses the challenge using the MediaPipe algorithm for real-time face landmark detection and tracking. Detected facial points can be recorded for analysis and for training machine learning models. The code runs across multiple platforms, including PCs, Nvidia Jetson, and Raspberry Pi, supporting face tracking, landmark analysis, and precise head and eye movement based on detected facial data.

An earlier version of this system focused on face recognition but lacked continuous face tracking and object-tracking capabilities. It can be accessed here for reference: <https://www.electronicsforu.com/electronics-projects/smart-robot-face-recognition>

MediaPipe can also detect and track common objects such as phones, bottles, and cups in real

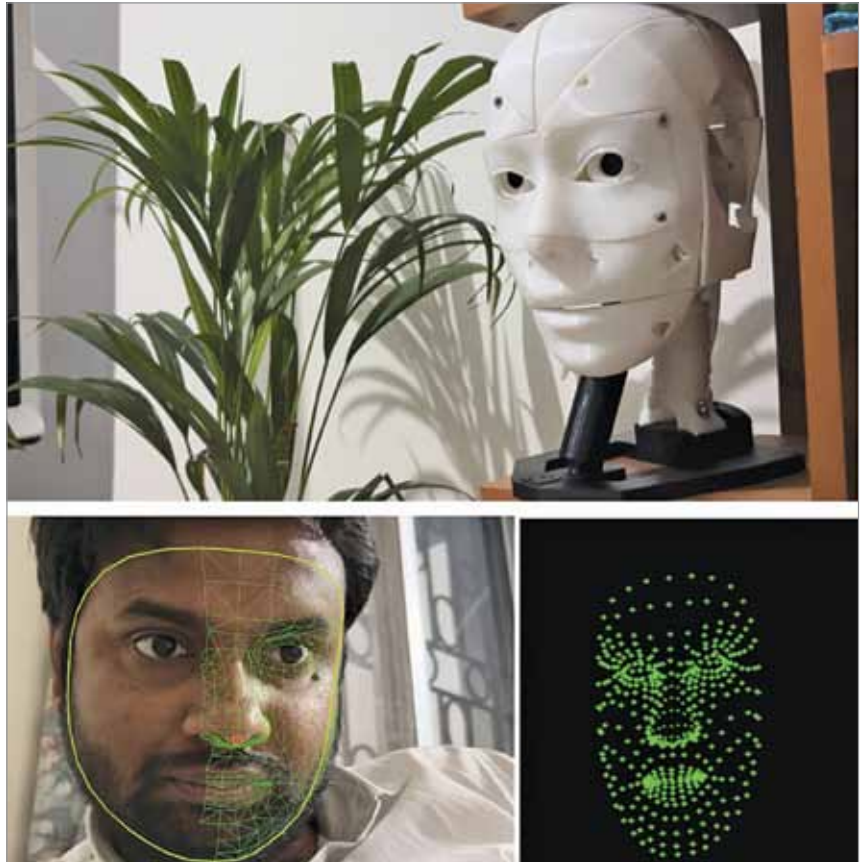


Fig. 1: Author's prototype robot tracking a face and detecting facial landmark points in real time

BILL OF MATERIALS

Name	Description	Quantity
3D printed parts of a robot	3D parts of the inMoover robot face	1
Servo motor	For eye movement and head movement: 2 micro geared, 2 metallic geared MG995 or similar	4
Raspberry Pi/Nvidia Jetson/Nvidia Orion	SBC for AI image processing and control	1
2A, 5V DC power supply	For robot and servo motor power	1
Raspberry Pi camera	5MP or 10MP CSI camera	1

EFY note: In addition to the components mentioned above, screws, a screwdriver, and similar hardware tools may be required to assemble the robot.

time. The robot can move its eyes and head to follow these objects while interacting with both people and its surroundings.

This capability enables the development of task-oriented robots

that can perform activities such as picking, sorting, delivering, or handing over objects. By combining facial landmark tracking with object detection, the robot becomes more aware and responsive. It